most basic conservation law associated to a closed system (the conservation of energy) as it is formulated in terms of the energy function. Second, it is the starting point for the quantum mechanical treatment of a system of particles, since the Schrödinger equation is written in terms of the quantum analogue of the energy function. Third, we may view the Hamiltonian as the *Legendre transformation* of the Lagrangian viewed as a function of the velocities. This is a *conjugacy* relationship which ensures that the properties of the two systems are, under typical conditions, equivalent.

Abstractly, a Legendre transformation of a given convex function $g = g(\boldsymbol{\xi}) : \mathbb{R}^m \to \mathbb{R}$ is a new function $\tilde{g} = \tilde{g}(\boldsymbol{\eta}) : \mathbb{R}^m \to \mathbb{R}$ defined by

$$\tilde{g}(\boldsymbol{\eta}) = \sup_{\boldsymbol{\xi}} (\boldsymbol{\eta}^T \boldsymbol{\xi} - g(\boldsymbol{\xi})),$$

where the symbol $\sup_{\boldsymbol{\xi}}$ means to find the supremum of the given expression with respect to variation of the vector $\boldsymbol{\xi}$. In the case of the Lagrangian $L(\boldsymbol{q}, \boldsymbol{v}) = \boldsymbol{v}^T \boldsymbol{M}(\boldsymbol{q})\boldsymbol{v}/2 - U(\boldsymbol{q})$ (which we have seen is the formulation of a mechanical system in generalized coordinates) we find

$$\sup_{\boldsymbol{v}} (\boldsymbol{p}^T \boldsymbol{v} - (\boldsymbol{v}^T \boldsymbol{M}(\boldsymbol{q})\boldsymbol{v}/2 - U(\boldsymbol{q}))) = \sup_{\boldsymbol{v}} (\boldsymbol{p}^T \boldsymbol{v} - \boldsymbol{v}^T \boldsymbol{M}(\boldsymbol{q})\boldsymbol{v}/2) + U(\boldsymbol{q}).$$

If $\boldsymbol{M}(\boldsymbol{q})$ is invertible, the supremum is achieved precisely when

$$\boldsymbol{v} = \boldsymbol{M}^{-1}(\boldsymbol{q})\boldsymbol{p},$$

which gives the standard definition of the momentum vector $\boldsymbol{p}$ in terms of velocities,

$$\boldsymbol{p} = \frac{\partial L}{\partial \dot{\boldsymbol{q}}} = \boldsymbol{M}(\boldsymbol{q})\dot{\boldsymbol{q}},$$

and, with $\boldsymbol{v} = \boldsymbol{M}^{-1}(\boldsymbol{q})\boldsymbol{p}$, the Legendre transformation results in a new function

$$H(\boldsymbol{q}, \boldsymbol{p}) \stackrel{\text{def}}{=} \boldsymbol{p}^T \boldsymbol{M}(\boldsymbol{q})^{-1} \boldsymbol{p}/2 + U(\boldsymbol{q}).$$

This is precisely the energy function, written in terms of positions and momenta. In this form it is termed the *Hamiltonian* and it provides an alternative description of the system. The equations of motion can be written

$$\dot{\boldsymbol{q}} = \frac{\partial H}{\partial \boldsymbol{p}}, \qquad \dot{\boldsymbol{p}} = -\frac{\partial H}{\partial \boldsymbol{q}}.$$

For constant $\boldsymbol{M}$ we obtain the dynamical equations,

$$\dot{\boldsymbol{q}} = \boldsymbol{M}^{-1}\boldsymbol{p}, \qquad \dot{\boldsymbol{p}} = \boldsymbol{F} = -\partial U/\partial \boldsymbol{q}.$$